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• Book suggestion:

Reference - 1st vol (Ronald...)

লেখক,

Arifur Haidar (জানগুরু প্রণালী) → যদি টেকনিক্যাল-
hard হয়

{Next class এর শুরু}

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(*) Mathematical Induction:

Ex! $1 + 2 + 2^2 + 2^3 + \dots + 2^n = 2^{n+1} - 1$

Basic Step:

$$2^{0+1} - 1 = 1$$

$$1 + 2 + 3 = 6 \quad (+4)$$

$$1 + 2 + 3 + 4 = 10$$

Induction part:

$$1 + 2 + 2^2 + \dots + 2^m = 2^{m+1} - 1 \quad \text{--- (ii)}$$

$$1 + 2 + 2^2 + \dots + 2^m + 2^{m+1} = 2^{m+2} - 1 \quad \text{--- (iii)}$$

(iii) গুরুত্বপূর্ণ last term এর ক্ষেত্রে (ii) এর দুইপাশে

(মাত্র একটি (iii) প্রাপ্ত Expression কে মনে রাখা)

$$(*) \quad 1 + 3 + 5 + \dots + (2n-1) = n^2 \quad \text{--- (i)}$$

Basic step:

Let assume the eqⁿ is true for $n=1$

Thus, ~~LS~~ $LS = 1$
 $RS = 1^2 = 1$

Inductive step: Let assume an integer m . and eqⁿ (i) is true for $n=m$

$$\therefore 1 + 3 + 5 + \dots + (2m-1) = m^2 \quad \text{--- (ii)}$$

Again, assume the consecutive number of m is $(m+1)$. So the eqⁿ (i) for $n=m+1$ is,

$$1 + 3 + 5 + \dots + (2m+1) = (m+1)^2 \quad \text{--- (iii)}$$

Now, add $(2m+1)$ to the both sides of eqⁿ (ii) And we get

$$1+3+5+\dots+(2m-1)+(2m+1)=m^2+2m+1$$

$$\text{or, } 1+3+5+\dots+(2m+1) = \cancel{(m+1)^2} \quad \boxed{(m+1)^2}$$

$$(1+1+1) \dots (1+1) = m^2 + 2 \cdot m \cdot 1 + 1^2$$

$$\text{or, } 1+3+5+\dots+(2m+1) = (m+1)^2$$

So, the eqn (i) is true for any $n \in \mathbb{N}$.
(proved)

$$(*) \quad 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$(*) \quad 1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \dots + n(n+1) = \frac{n(n+1)(n+2)}{3}$$

[Hw: 2nd to math $\sum$]

$$(*) 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6} \quad (i)$$

Basic step:

Let, assume, the eqⁿ is true for $n=1$,

then,

$$LS = 1^2 = 1$$

$$RS = \frac{1(1+1)(2 \cdot 1+1)}{6}$$

$$= \frac{1 \cdot 2 \cdot 3}{6}$$

$$= 1$$

$$\therefore LS = RS$$

Induction step:

Let, assume an integer m . And

eqⁿ (i) is true for $n=m$

then,

$$1^2 + 2^2 + 3^2 + \dots + m^2 = \frac{m(m+1)(2m+1)}{6} \quad (ii)$$

Again, assume the consecutive number of m is $(m+1)$. So the eqⁿ (i) for

$n = m+1$ is,

$$1^2 + 2^2 + 3^2 + \dots + (m+1)^2 = \frac{(m+1)(m+2)(2m+3)}{6} \quad \dots \dots (ii)$$

Now,

add $(m+1)^2$ to the both sides of

(ii) we get,

$$1^2 + 2^2 + 3^2 + \dots + m^2 + (m+1)^2 = \frac{(m+1)(m+1+1)}{6} \{2(m+1) + 1\}$$

$$\begin{aligned}
& 1^2 + 2^2 + 3^2 + \dots + m^2 + (m+1)^2 \\
&= \frac{m(m+1)(2m+1)}{6} + (m+1)^2 \\
&= \frac{(m^2+m)(2m+1) + 6(m^2+2m+1)}{6} \\
&= \frac{2m^3 + m^2 + 2m^2 + m + 6m^2 + 12m + 6}{6} \\
&= \frac{2m^3 + 9m^2 + 13m + 6}{6} \\
&= \frac{2m^3 + 2m^2 + 7m^2 + 7m + 6m + 6}{6} \\
&= \frac{2m^2(m+1) + 7m(m+1) + 6(m+1)}{6} \\
&= \frac{(m+1)(2m^2 + 7m + 6)}{6} \\
&= \frac{(m+1)(2m^2 + 4m + 3m + 6)}{6} \\
&= \frac{(m+1)\{2m(m+2) + 3(m+2)\}}{6} \\
&= \frac{(m+1)(m+2)(2m+3)}{6}
\end{aligned}$$

So, the eqⁿ (1) is true for all $n \in \mathbb{N}$.